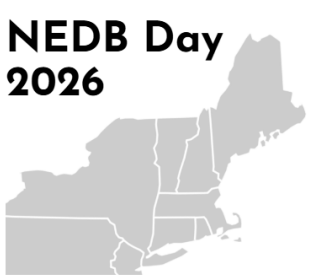


DiskJoin: Large-scale Vector Similarity Join with SSD

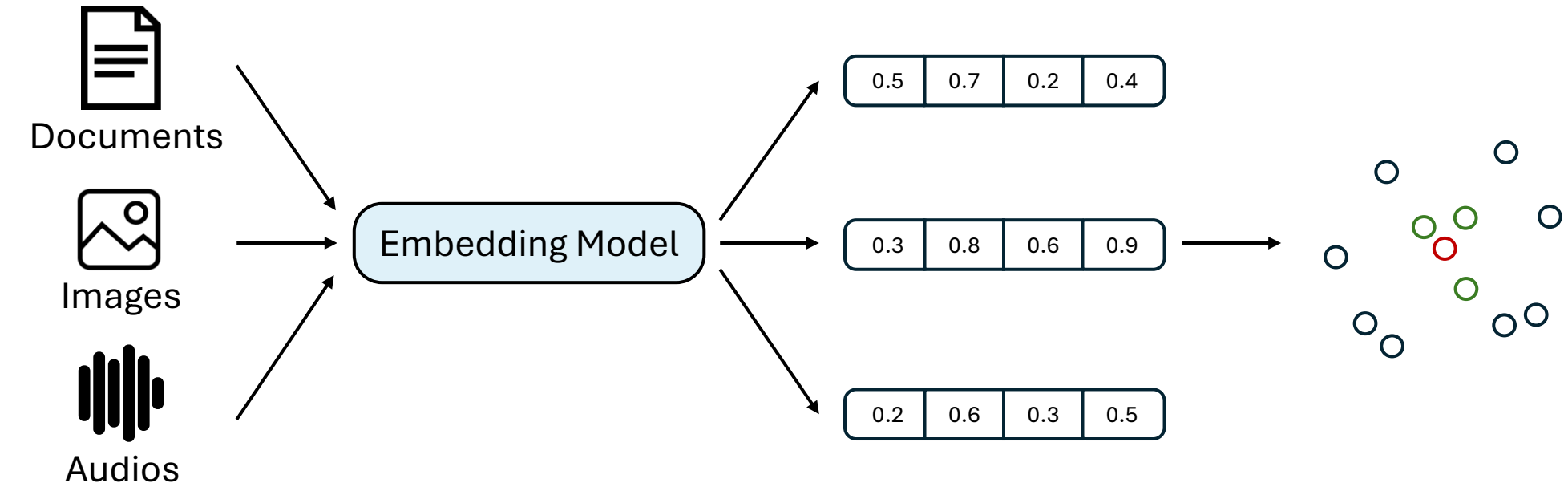
Yanqi Chen, Xiao Yan, Alexandra Meliou, Eric Lo

University of Massachusetts Amherst

1/16/2026

To be presented at SIGMOD 2026

Vector Embeddings and Similarity Join

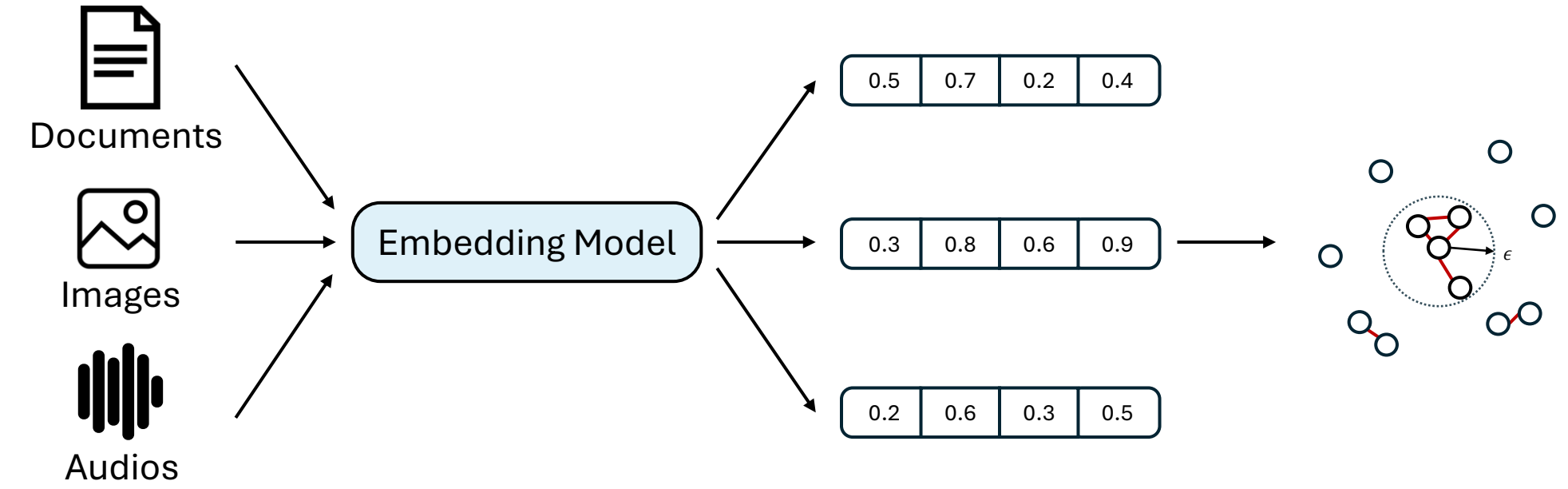


Data Objects

Vector Embeddings

Similarity Search

Vector Embeddings and Similarity Join

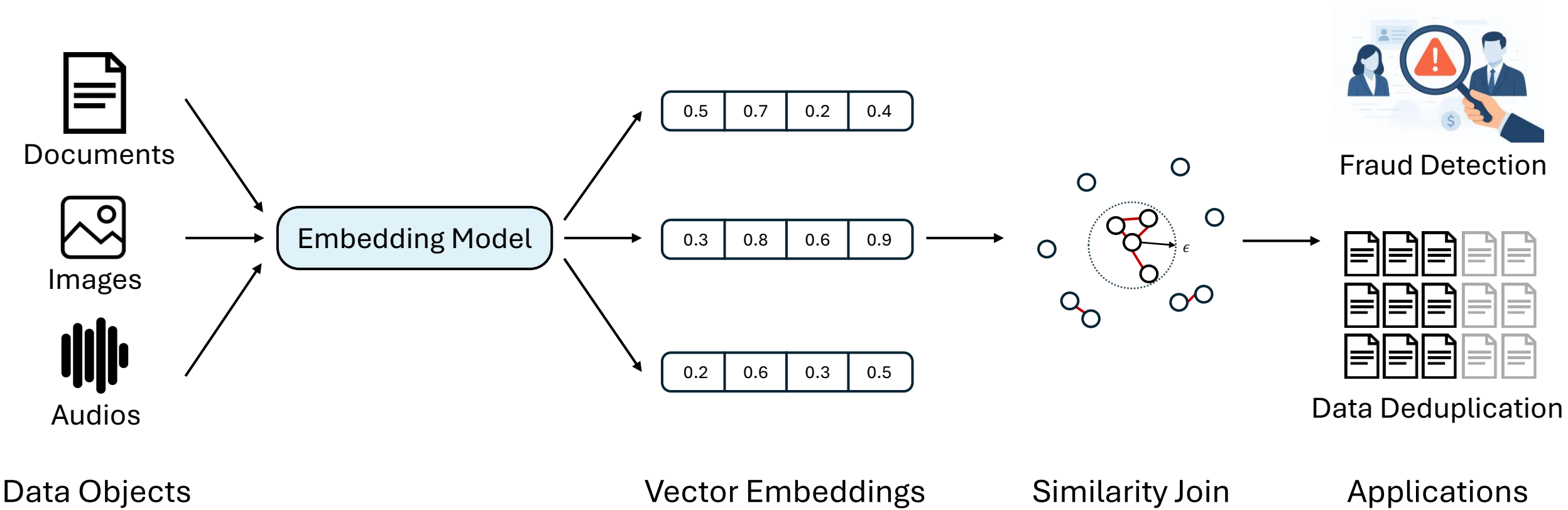


Data Objects

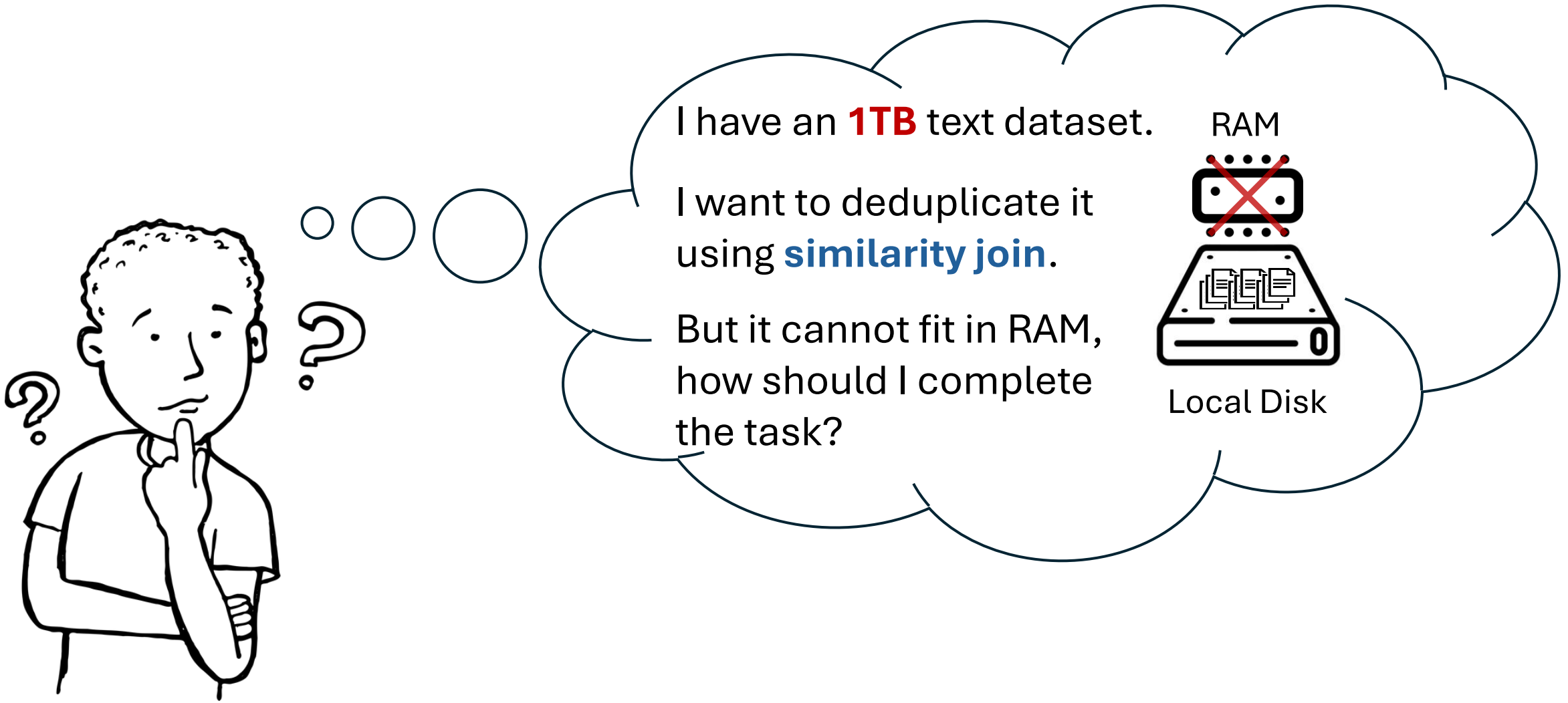
Vector Embeddings

Similarity Join

Vector Embeddings and Similarity Join



Scaling to Large Data Sizes



Scaling out: Distributed Methods

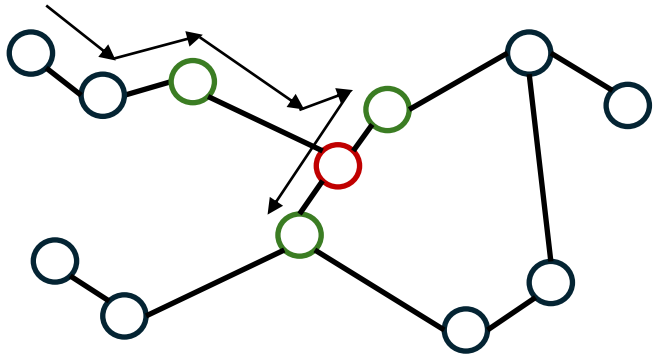
- Prior work has explored distributed methods
  - e.g., MAPSS, ClusterJoin, C2Net
  - Distribute dataset to multiple machines
  - Conduct computation via MapReduce

✗ Require tedious cluster setup

✗ High cost: CPU, RAM, disk for each machine

✗ Inefficient: high inter-machine communication cost

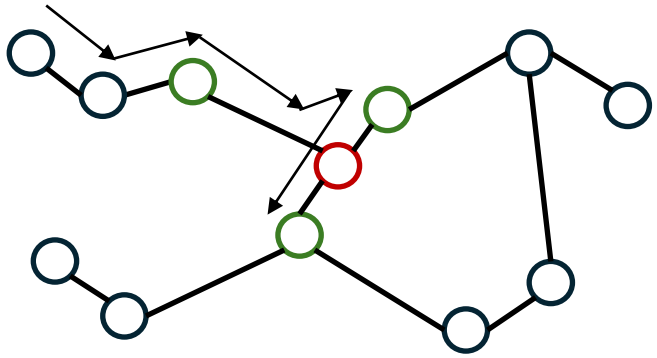
Similarity Join with SSD: Index Choice



Graph Index

DiskANN
[Subramanya et al., NeurIPS 2019]

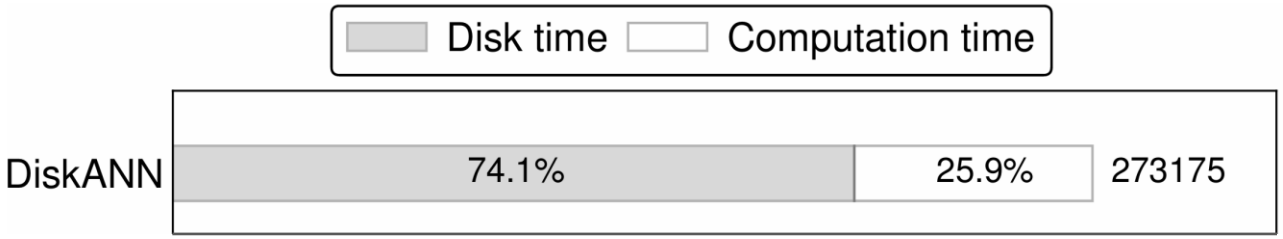
Similarity Join with SSD: Index Choice



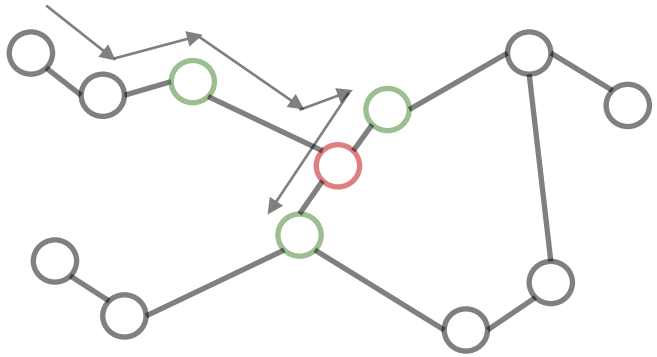
Graph Index

DiskANN
[Subramanya et al., NeurIPS 2019]

Read amplification!



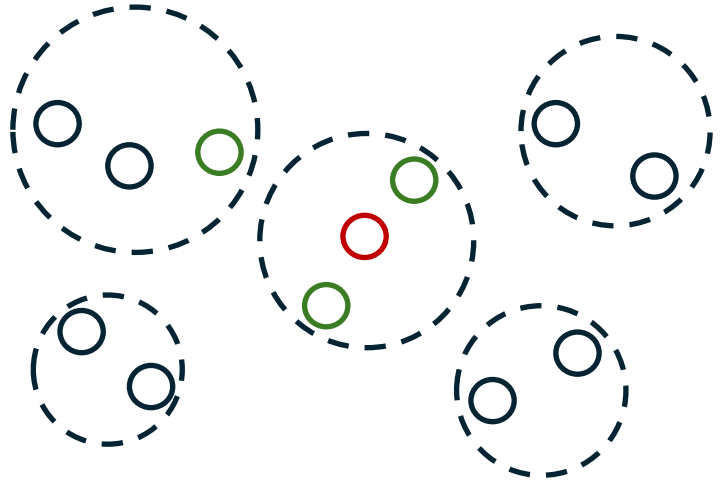
Similarity Join with SSD: Index Choice



Graph Index

DiskANN
[Subramanya et al., NeurIPS 2019]

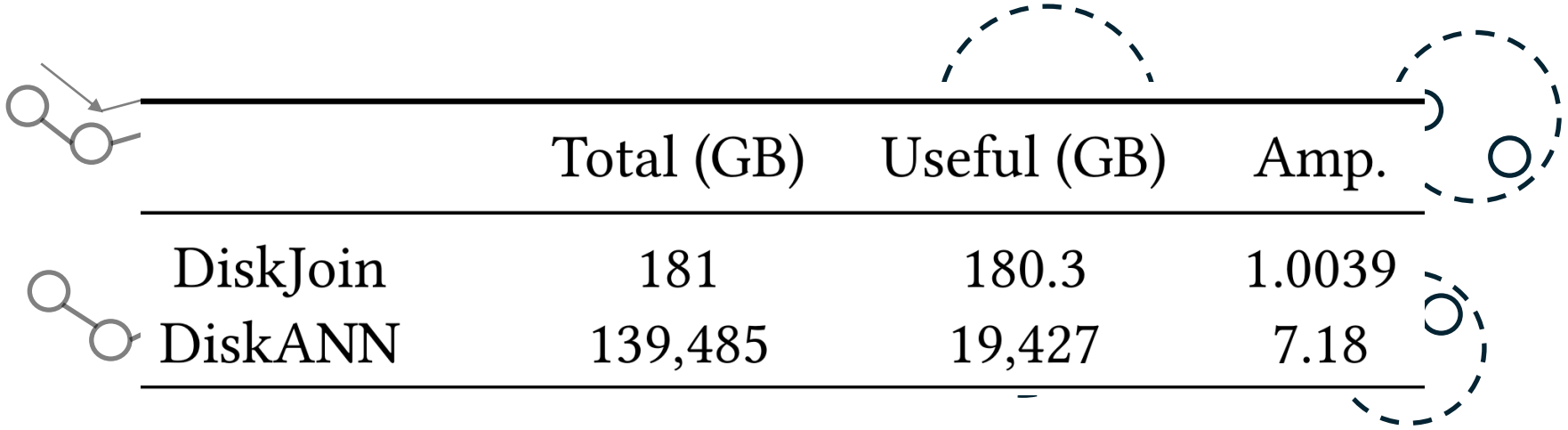
Read amplification!



Cluster Index

Read block of vectors ✓

Similarity Join with SSD: Index Choice



	Total (GB)	Useful (GB)	Amp.
DiskJoin	181	180.3	1.0039
DiskANN	139,485	19,427	7.18

Graph Index

DiskANN
[Subramanya et al., NeurIPS 2019]

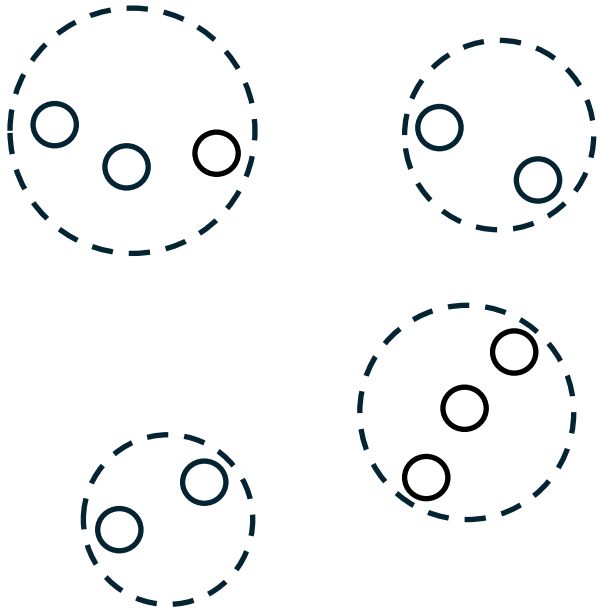
Read amplification!

Cluster Index

Read block of vectors ✓

Cluster-based Framework

1. Cluster the vectors
2. Identify candidate cluster pairs
3. For each candidate pair
    1. Load clusters from disk
    2. Compare vectors across clusters



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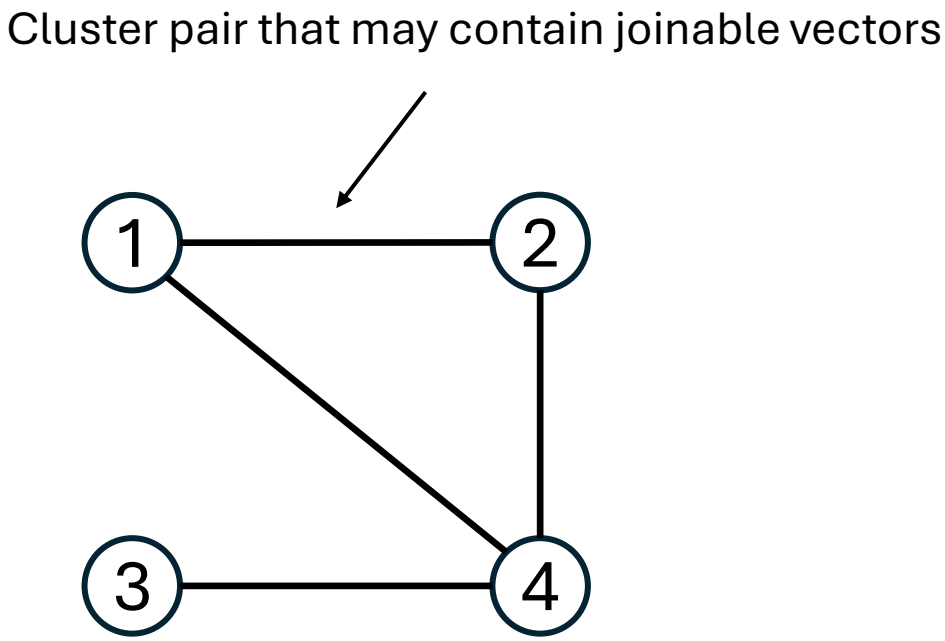

Cluster-based Framework

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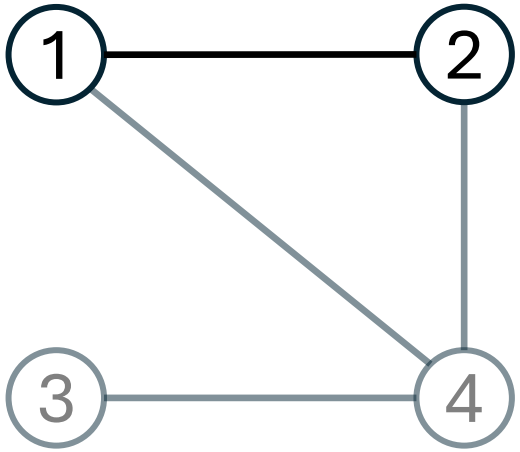
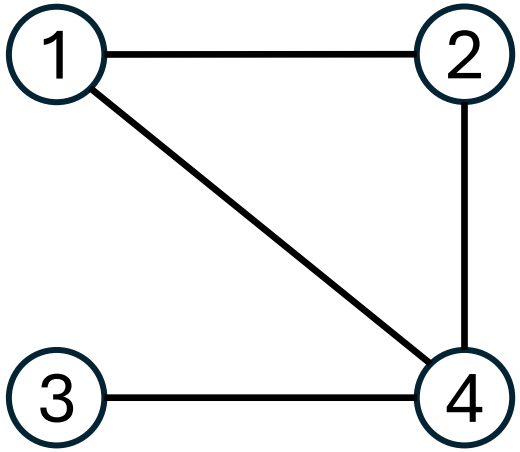
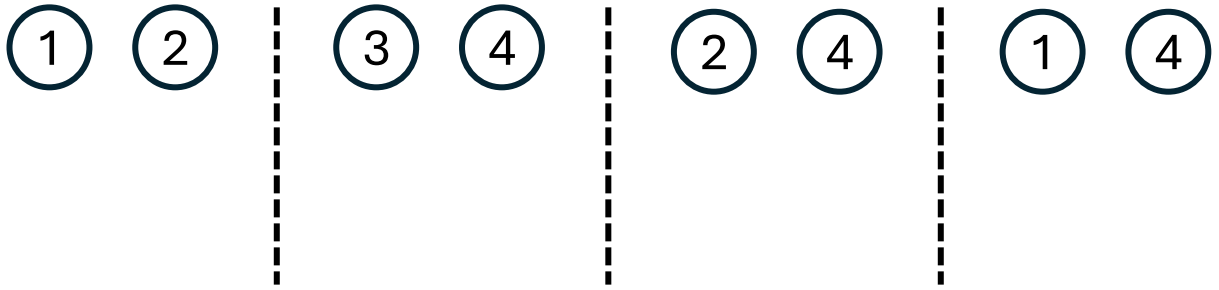


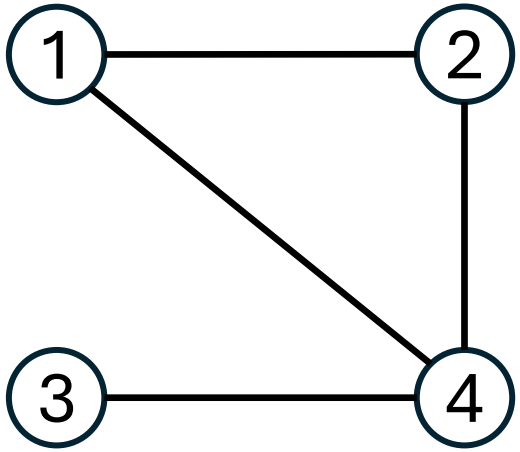
Repetitive Data Access

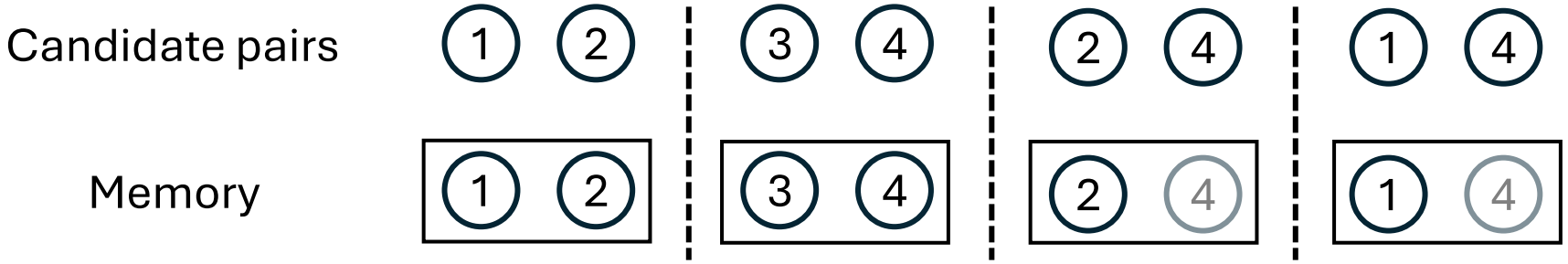


Candidate pairs



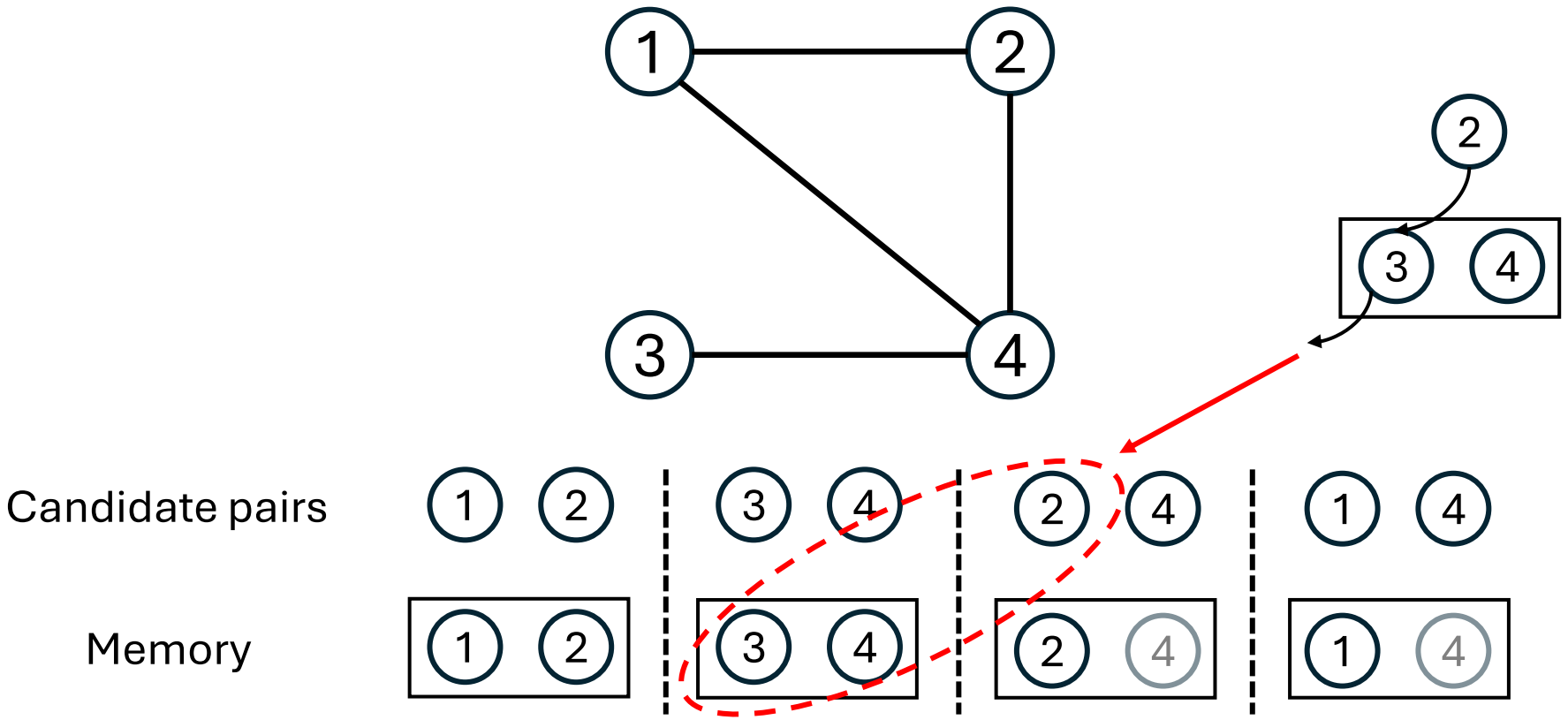
Data Reuse via Caching





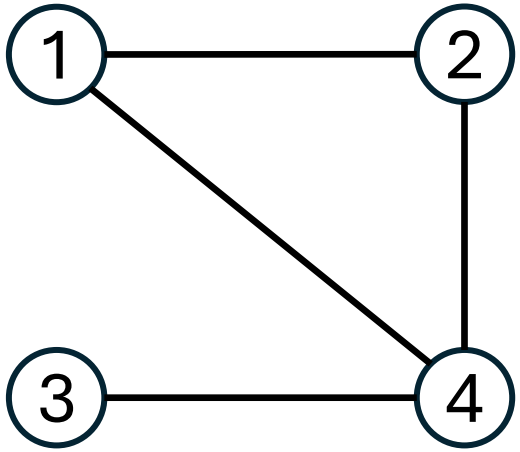
Disk read time -> Cache miss rate

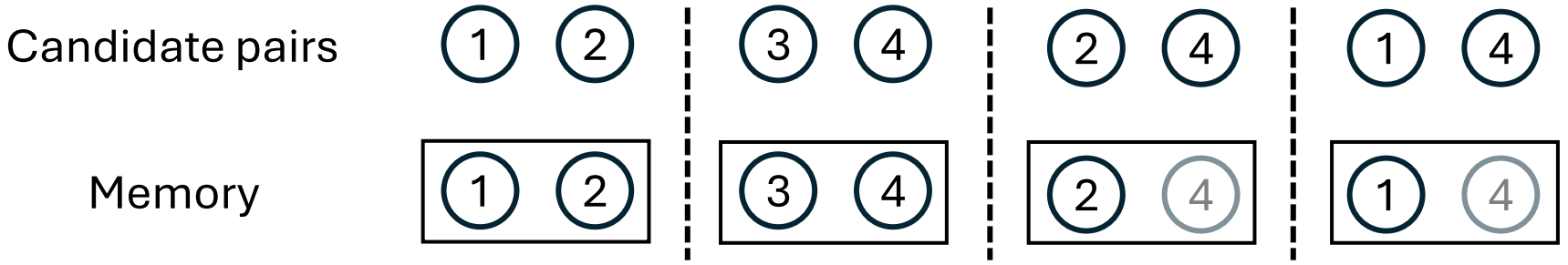
Data Reuse via Caching



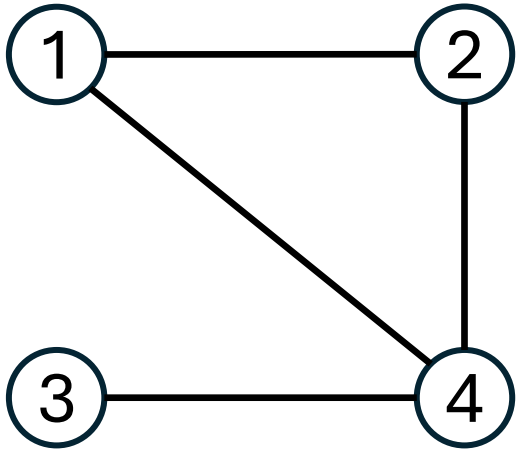
Which cluster to evict matters

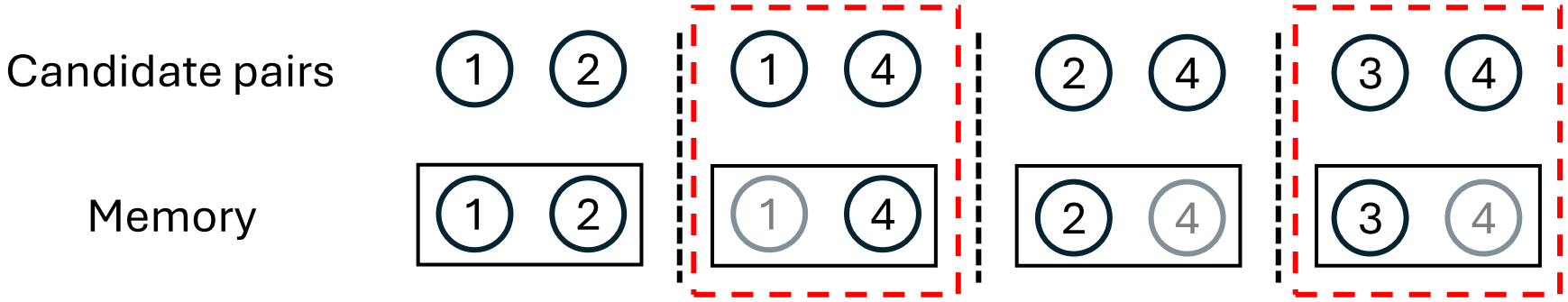
Data Reuse via Caching





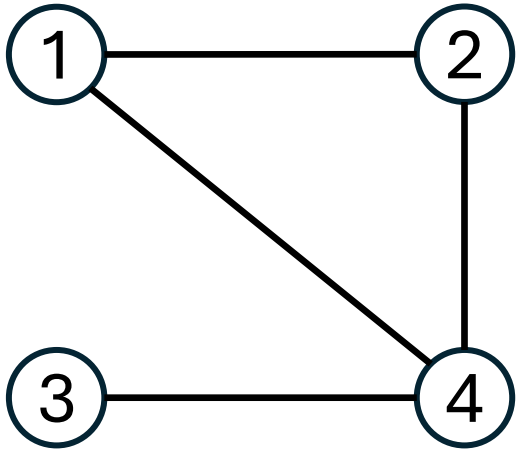
Data Reuse via Caching

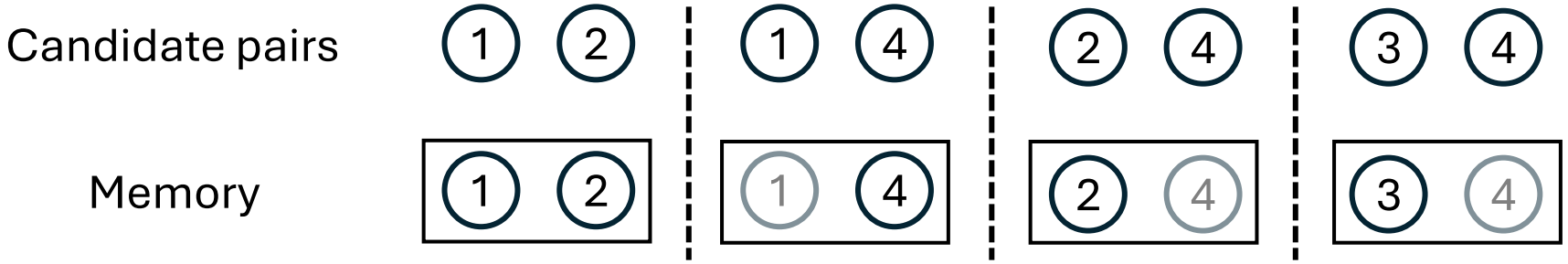




Processing order also matters

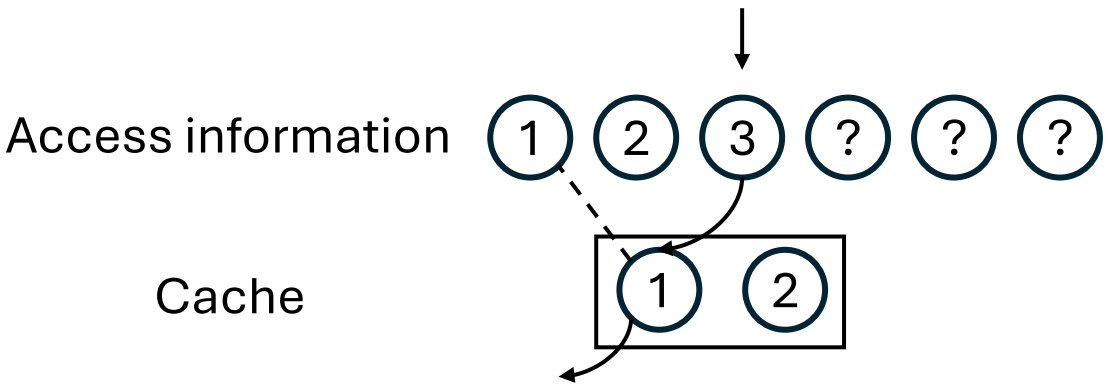
Data Reuse via Caching





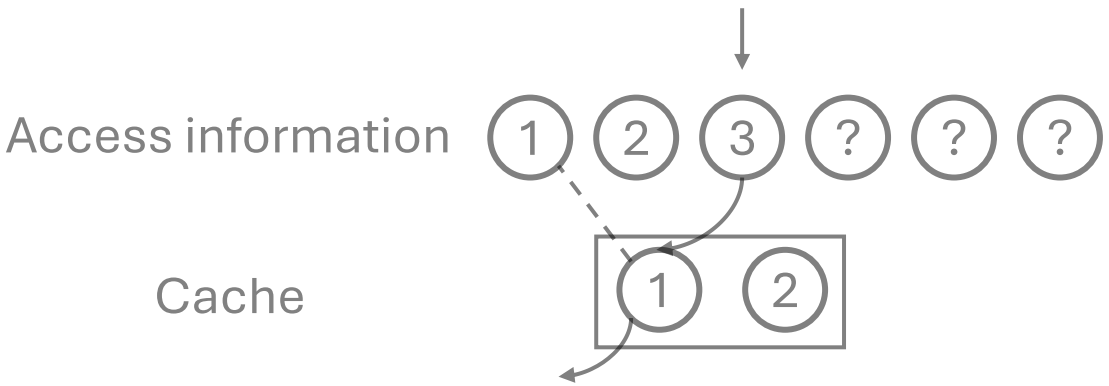
How to decide the optimal eviction policy and processing order?

Optimal Eviction Policy

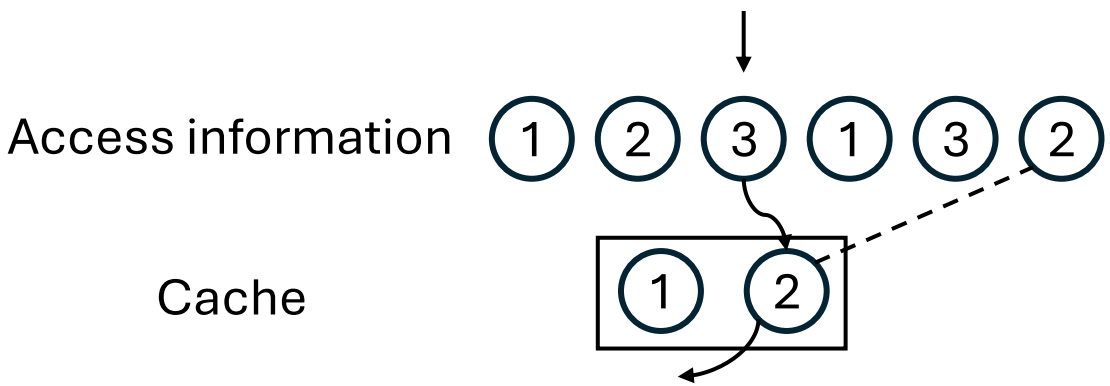


LRU: looks in past

Optimal Eviction Policy



LRU: looks in past

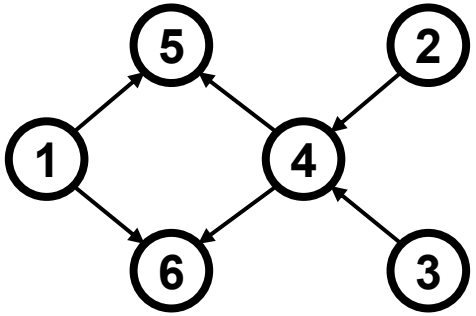


Belady's: looks in **future**

Provably optimal!

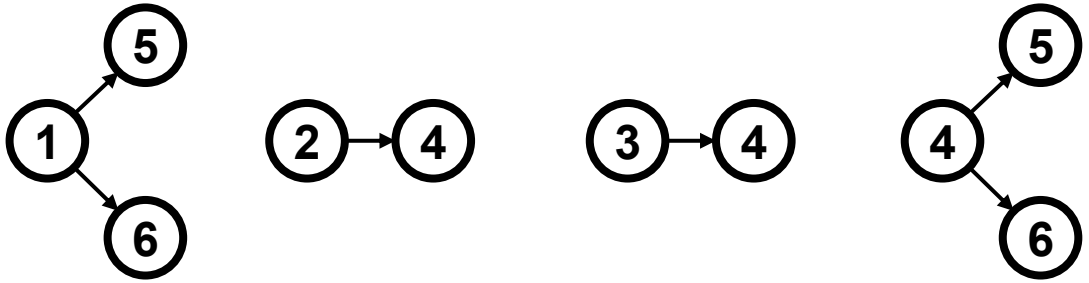
Heuristic for Ordering

Intuition 1: Group by node



dependency graph

2|E|

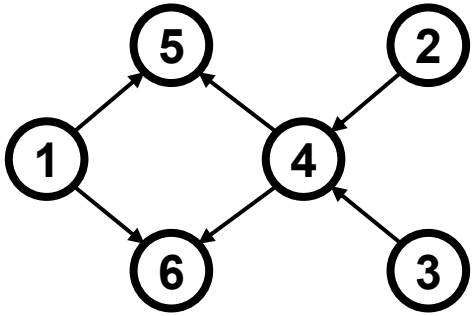


|V|+|E|

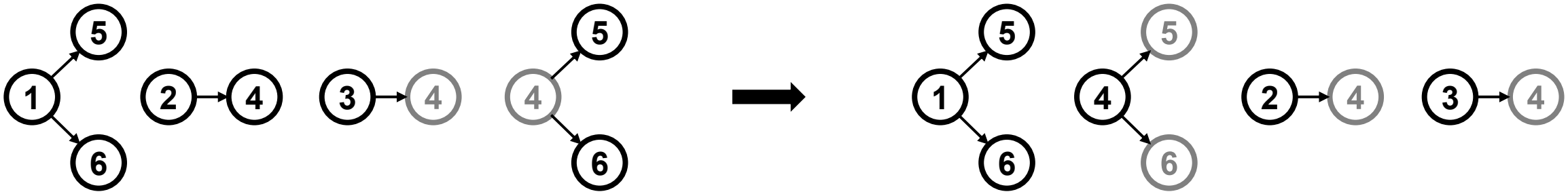
Heuristic for Ordering

Intuition 1: Group by node

Intuition 2: Place nodes with high neighbor overlap together



dependency graph



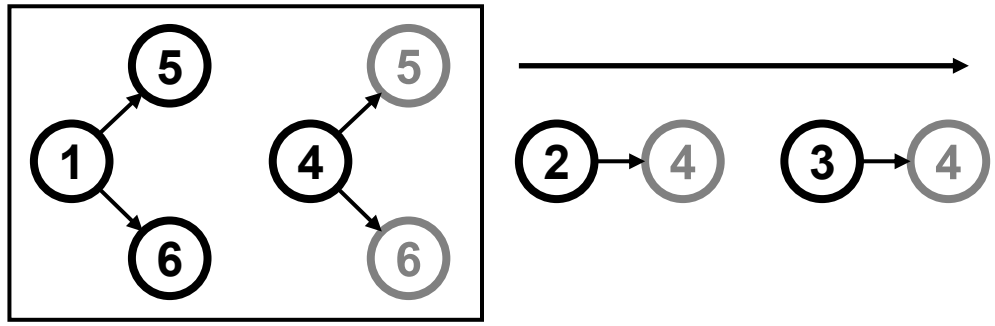
Processing Order Optimization

$$maximize \quad F(P) = \sum_{0<P(v)-P(u)\leq w} |N(u) \cap N(v)|$$

Sliding window — Neighbor overlap

$w = 2$

$F(P) = 2 + 1 + 1 = 4$

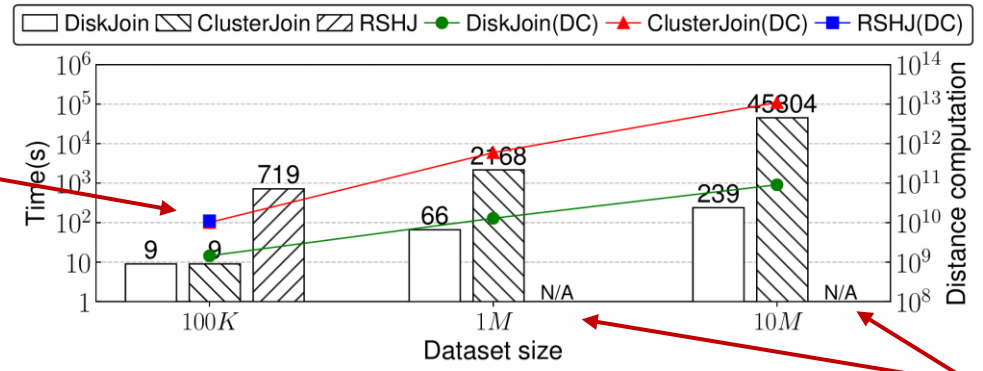


Greedy [Wei et al., SIGMOD 2016]: $P[i] = \underset{v}{argmax} \sum_{j=i-w}^{i-1} |N(P[j]) \cap N(v)|$

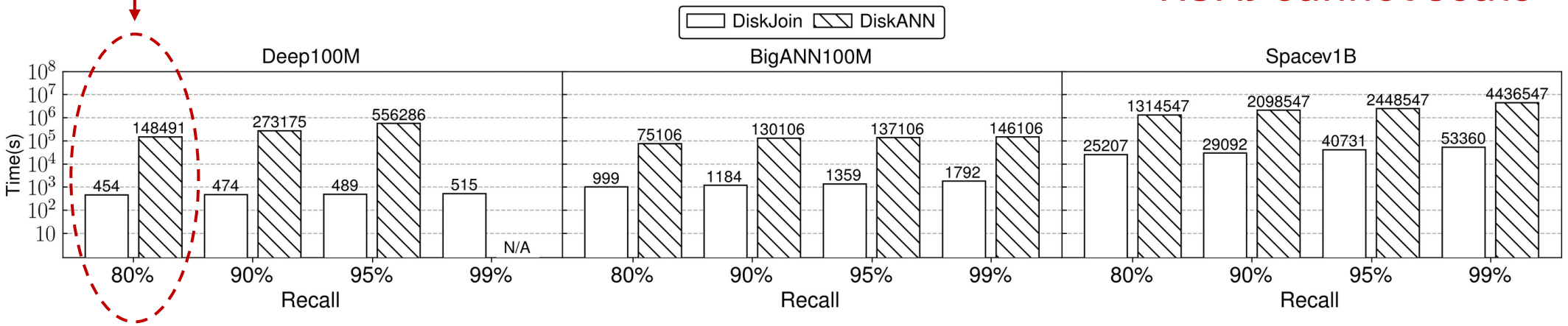
Experiments

- DiskJoin outperforms distributed methods and DiskANN by orders of magnitude

DiskJoin is 2-3 orders of magnitude faster

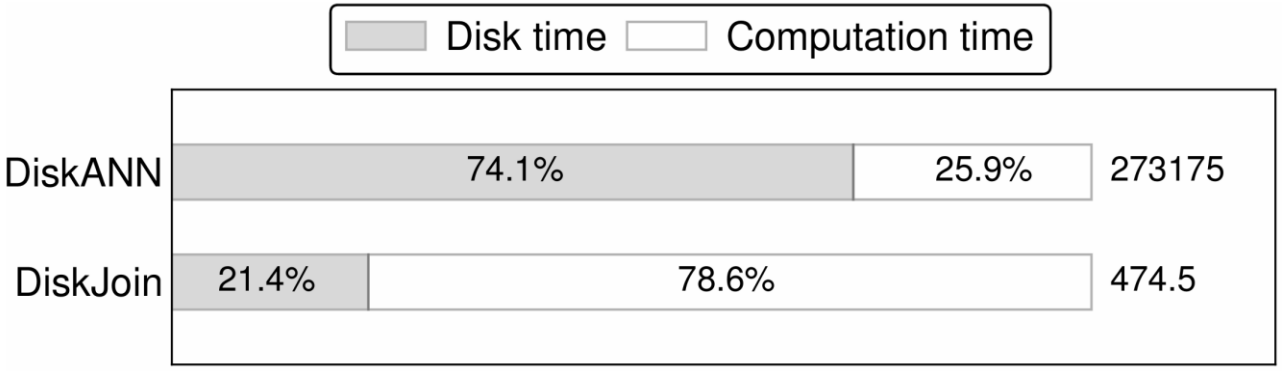


RSHJ cannot scale



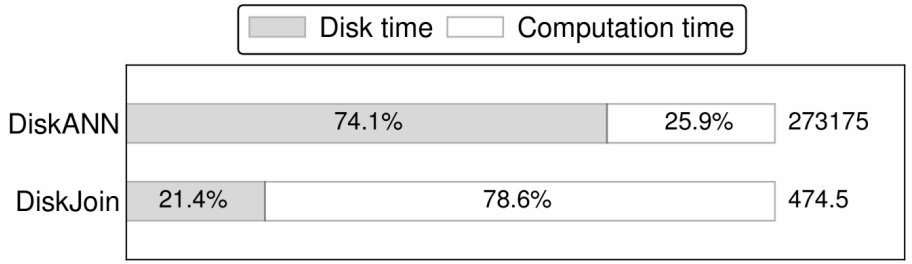
Experiments

- DiskJoin successfully eliminates the disk access time bottleneck

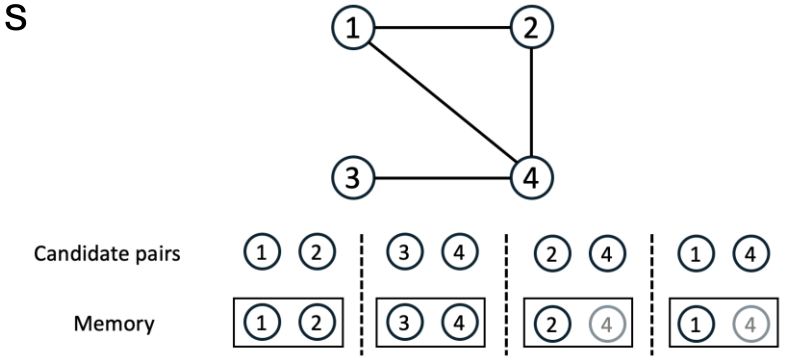


Summary

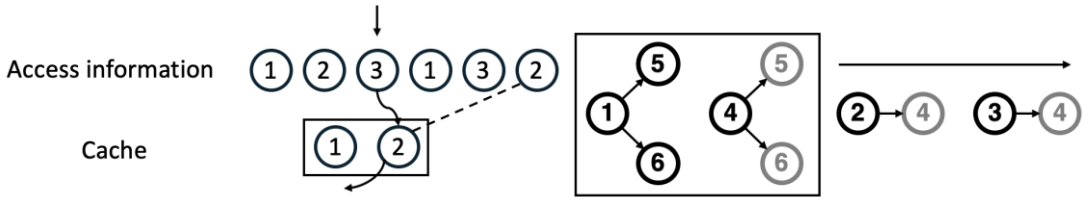
- Identify the disk bottleneck of disk similarity join due to read amplification and repetitive data access.



- Based on cluster-based framework, introduce memory caching to achieve data reus.



- Minimize cache miss rate by optimal cache management with Belady's algorithm and task ordering.



- Evaluation results show DiskJoin outperforms competitors by orders of magnitude.